

## Campus Quest - Test Report

| Test                     | Setup / Action                                 | Predicted Result                         | Actual Result                                            | Pass/Fail   | Defect Exposable                               |
|--------------------------|------------------------------------------------|------------------------------------------|----------------------------------------------------------|-------------|------------------------------------------------|
| <b>T1 Empty</b>          | Show leaderboard with size 0                   | No read; clear "no teams" output.        | "Total: 0 teams" printed.                                | <b>Pass</b> | Uninitialized pointer crash.                   |
| <b>T2 First add</b>      | Add ID 101                                     | Size becomes 1; record is visible.       | Record added successfully.                               | <b>Pass</b> | Failed memory allocation on first add.         |
| <b>T4 Duplicate</b>      | Add ID 101 again                               | Rejected; size and records unchanged.    | Rejected with error message.                             | <b>Pass</b> | Data corruption from ID collisions.            |
| <b>T6 Search absent</b>  | Search unused ID                               | Report not found; no mutation.           | Rejected with error message.                             | <b>Pass</b> | Null pointer dereference or infinite loop.     |
| <b>T8 Sort identity</b>  | Load equal/different scores (110\ 2 vs 110\ 7) | Correct rank; names remain attached.     | Sorted correctly using tie-breaker.                      | <b>Pass</b> | Incorrect sorting logic or data swap error.    |
| <b>T9 Malformed file</b> | Add bad fields: a\ b\ c, -, 104\ ...\ (r\ 1    | Lines rejected; valid records unchanged. | Bad lines safely skipped. Extra trailing fields ignored. | <b>Pass</b> | Buffer overflow or start-up crash via parsing. |